

7.5 Carboxylic Acids & Derivatives (A Level Only)

Question Paper

Course	CIE A Level Chemistry
Section	7. Organic Chemistry (A Level Only)
Topic	7.5 Carboxylic Acids & Derivatives (A Level Only)
Difficulty	Easy

Time allowed: 30

Score: /18

Percentage: /100

Question 1a

This question is about derivatives of carboxylic acids.

Fig 1.1 shows the structure of a carboxylic acid derivative.

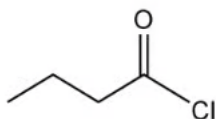


Fig 1.1

State the name of the carboxylic acid derivative in Fig 1.1.

[1 mark]

Question 1b

Ethanoyl chloride reacts with ethanol to form ethyl ethanoate as shown in Fig 1.2.

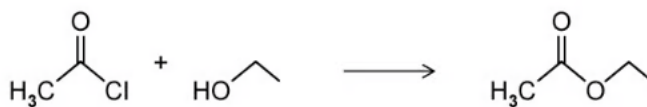


Fig 1.2

i)

Name the other product formed in this reaction.

[1]

ii)

This reaction can be described as an esterification reaction or an acylation reaction.

Explain why this reaction can also be described as a condensation reaction.

[1]

[2 marks]

Question 1c

Ethanoyl chloride can also react with ammonia to form ethanamide.

Write an equation for this reaction.

[1 mark]

Question 1d

During the preparation of propanoyl chloride, SO_2 and HCl were also produced.

Name the two reactants used to form propanoyl chloride.

[1 mark]

Question 2a

Benzoic acid can be formed from the oxidation of methylbenzene.

Draw the displayed formula of benzoic acid.

[1 mark]

Question 2b

i)

State the reagents and conditions needed to oxidise methylbenzene to benzoic acid.

[3]

ii)

Give **two** observations that could be made during this reaction.

[2]

[5 marks]

Question 2c

Write an equation to show that benzoic acid acts as a weak acid.

[1 mark]

Question 3a

Propanoyl chloride is used for the synthesis of other organic compounds.

Draw the skeletal formula for propanoyl chloride.

[1 mark]

Question 3b

i)

Name the two products formed when propanoyl chloride undergoes hydrolysis.

[2]

ii)

Name the mechanism by which hydrolysis occurs.

[1]

[3 marks]

Question 3c

Explain why propanoyl chloride will more readily undergo hydrolysis compared to 2-chloropropane.

[2 marks]

Model Answers: Easy

1a

a)

i) The carboxylic acid derivative in Fig 1.1 is:

- Butanoyl chloride; [1 mark]

[Total: 1 mark]

- Fig 1.1 is an acyl chloride / acid chloride
 - This means that you identify the longest carbon chain and add *-oyl chloride* to the chain name

1b

b)

i) The other product formed in this reaction is:

- Hydrogen chloride / HCl; [1 mark]

ii) This reaction can also be described as a condensation reaction because:

- A small molecule / hydrogen chloride / HCl is eliminated; [1 mark]

[Total: 2 marks]

- Acyl chlorides can react with alcohols to form esters
- This is just like the reaction of carboxylic acids with alcohols
- The only difference is the other product
 - Carboxylic acids form esters and water
 - Acyl chlorides form esters and hydrogen chloride
- Some students mistake a condensation reaction as one in which water is produced, but it is one in which a small molecule is produced

1c

c) The equation for this reaction is:

- $\text{CH}_3\text{COCl} + \text{NH}_3 \rightarrow \text{CH}_3\text{CONH}_2 + \text{HCl}$
OR
- $\text{CH}_3\text{COCl} + 2\text{NH}_3 \rightarrow \text{CH}_3\text{CONH}_2 + \text{NH}_4\text{Cl}$; [1 mark]

[Total: 1 mark]

- When ethanoyl chloride and ammonia react, the amide and HCl are formed



- The ammonia is basic so will react with the HCl to form the ammonium salt NH_4Cl



- Either equation is accepted but the first equation would be the answer given by most students

1d

d) The two reactants are:

- Propanoic acid;
AND
Sulfur dichloride oxide / SOCl_2 ; [1 mark]

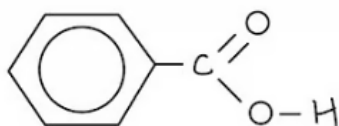
[Total: 1 mark]

- The equation for the reaction is:



2a

a) The displayed formula of benzoic acid is:



- Correct structure showing all bonds; [1 mark]

[Total: 1 mark]

- **Remember:** For displayed formulae, all bonds should be shown, including those between the O and H atoms

2b

b)

i) The reagents and conditions needed to oxidise methylbenzene to benzoic acid are:

- Hot alkaline KMnO_4 / potassium manganate (VII); [1 mark]
- Dilute acid / dilute sulfuric acid; [1 mark]
- Heat **AND** reflux; [1 mark]

ii) Two observations that could be made during this reaction are:

- Solution loses purple colour; [1 mark]
- Brown precipitate (of MnO_2); [1 mark]

[Total 5 marks]

- You should be aware of potassium manganate as an oxidising agent in the context of oxidising alcohols
 - Here, the KMnO_4 changes from purple to a colourless solution due to the manganese ions being reduced from Mn^{7+} to Mn^{2+}
- For the oxidation of methylbenzene, the purple colour is replaced by a brown precipitate of MnO_2 because the manganese ions have been reduced from Mn^{7+} to Mn^{4+}
- You must make sure you have indicated that it is alkaline KMnO_4 , as it has had sodium carbonate added to it

2c

c) An equation to show that benzoic acid acts as a weak acid is:

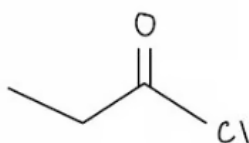
- $\text{C}_6\text{H}_5\text{COOH}(\text{aq}) \rightleftharpoons \text{C}_6\text{H}_5\text{COO}^-(\text{aq}) + \text{H}^+(\text{aq})$; [1 mark]

[Total: 1 mark]

- Weak acids only partially ionise in water
- The position of the equilibrium lies to the left so there is a lower concentration of H^+ ions
- Don't forget to include the symbol for a reversible reaction

3a

a) The skeletal formula for propanoyl chloride is:



- Correct skeletal formula; [1 mark]

[Total: 1 mark]

- You must be able to draw the skeletal or displayed formulae for any given compound

3b

b)

i) The two products formed when propanoyl chloride undergoes hydrolysis are:

- Propanoic acid; [1 mark]

i) The two products formed when propanoyl chloride undergoes hydrolysis are:

- Propanoic acid; [1 mark]
- HCl; [1 mark]

ii) The mechanism by which hydrolysis occurs is:

- Addition-elimination; [1 mark]

[Total: 3 marks]

- The hydrolysis of an acyl chloride will form the corresponding carboxylic acid and HCl
- The water molecule acts as a nucleophile and attacks the carbonyl carbon atom
- The HCl molecule is then eliminated
- For harder questions, you must be able to draw the mechanism for this reaction

3c

c) Propanoyl chloride will more readily undergo hydrolysis compared to 2-chloropropane because:

- It has a weaker C-Cl bond; [1 mark]
- Which can more easily break; [1 mark]

[Total: 2 marks]

- The question is only two marks so a detailed explanation is not required
- The strength of the C-Cl bond determines the tendency of these compounds to undergo hydrolysis
- 2-chloropropane is an alkyl chloride, propanoyl chloride is an acyl chloride
- The C-Cl bond is weaker in propanoyl chloride because the carbonyl carbon atom is bonded to oxygen - a very electronegative element which draws electron density away from the carbonyl carbon, leaving it with a slightly positive charge and therefore susceptible to attack by nucleophiles